

Digital Wind

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Table of Contents

1. Abstract	3
2. Introduction - The Offshore Acceleration Opportunity.....	3
3. Policy & Regulatory Context.....	4
4. Irish Power System Context & Fundamentals	5
4.1 AC System Fundamentals	6
4.2 System Non-Synchronous Penetration (SNSP)	6
4.3 Transmission Infrastructure.....	7
4.4 Interconnectors and Cross-Border Flows.....	7
5. Empirical Constraint Diagnosis and Curtailment Decomposition	8
5.1 Empirical System Baseline 2025 (Stage 1).....	8
5.2 Curtailment Decomposition and Constraint Attribution 2025 (Stage 2).....	10
6. Offshore Wind Expansion Stress Testing (Stage 3)	12
6.1 Offshore Dataset and Profile Construction	12
6.2 Spatial Transmission Injection Layer.....	13
6.3 Offshore Capacity Scenarios.....	13
6.4 Constraint Re-Evaluation Framework.....	14
6.5 Quantified Stress Results	15
6.6 Structural Interpretation	15
7. Predictive Digital Twin - Decision-Support Layer (Stage 4).....	15
7.1 Decision-State Formalisation	16
7.2 Engineered Constraint Indicators.....	16
7.3 Multi-Horizon Forecast Framework	17
7.4 Multi-model Machine Learning Architecture	18
7.5 Closed-Loop Digital Twin Simulation	20
8. Results & Quantified Impact	22
8.1 Predictive Model Performance Summary	22
8.2 Closed-Loop Mitigation Results	23
8.3 Structural Interpretation	23
9. Feasibility & Implementation Study	23
9.2 Data Requirements.....	24
9.3 Operational Integration Pathway	24
9.4 Scalability and Extension.....	24
10. Strategic Implementation Roadmap & Future Development.....	25
11. Conclusion	26
12. References.....	26
13. Appendices.....	28

13.1 Appendix A - AI Use Declaration	28
13.2 Appendix B - Team Contribution Statement	29
13.3 Appendix C - Reproducibility Statement	29

1. Abstract

Ireland's 2030 offshore wind target introduces structural integration pressure within a stability-constrained transmission network. As renewable penetration increases, transmission congestion, SNSP ceilings, and interconnector scheduling constraints generate structural curtailment [1].

However, expanding wind capacity does not automatically translate into increased usable energy. As renewable penetration rises, operational constraints - including System Non-Synchronous Penetration (SNSP) limits, transmission bottlenecks, and interconnector saturation, where cross-border flows are partially pre-committed through day-ahead and intraday market decisions - can lead to structural curtailment and wasted wind energy [3] [4].

This project presents *Digital Wind*, a grid-informed digital twin developed and implemented as an initial operational prototype using the Irish grid's system data to diagnose, stress-test, and optimise offshore wind integration under real-world operational constraints.

Using 2025 quarter-hourly system data covering Ireland (IE), Northern Ireland (NI), and the All-Island (AI) network [5], the framework establishes a system-wide baseline of grid behaviour. It quantifies demand patterns, renewable generation dynamics, SNSP operation, and the scale and distribution of dispatch down, providing a foundational view of system performance under real operational conditions [2].

The analysis first separates observed curtailment into transmission-driven congestion effects and system-wide SNSP stability constraints. It then introduces synthetic offshore wind scenarios (0.5 GW, 1 GW, and 5 GW), modelled using realistic capacity factors and geographically integrated into the existing Irish transmission network through location-aware assumptions.

The prototype subsequently implements a predictive decision-support layer combining probabilistic curtailment risk modelling, conditional interconnector saturation forecasting under forecast net-demand and market-scheduled flow conditions, and magnitude estimation to enable pre-emptive export and storage coordination prior to day-ahead and intraday market gate closure [4].

Under an offshore expansion stress scenario aligned with Ireland's 2030 targets [8], the implemented system reduces curtailed wind energy without requiring additional physical infrastructure. Across scenarios, average curtailment mitigation reaches 10.7% under intraday (4-hour) control and 9.5% under day-ahead (24-hour) control, demonstrating measurable gains from shorter-horizon predictive coordination within existing Irish grid constraints.

2. Introduction - The Offshore Acceleration Opportunity

Ireland's progress to date reflects significant operational and infrastructural advancement. The all-island system, operated by EirGrid and SONI, already integrates among the highest levels of non-synchronous generation globally [1]. In December 2025 alone, renewables supplied 44% of electricity demand, with wind contributing 39% and battery discharge reaching record levels of 396 MW [2]. Through sustained system services reform and disciplined operational management, the SNSP limit has been progressively raised while maintaining security of supply on a relatively small and electrically isolated AC grid [1].

Grid development is continuing in parallel with renewable expansion. In 2025, fourteen grid upgrade projects were energised, including major reinforcements at Corduff 220 kV Station in North Dublin and upgrades across multiple 110 kV circuits such as Glenree–Moy and Flagford–Sliabh Bawn [6]. The completion of works at Derryiron 110 kV Station has unlocked additional renewable connection capacity [6], while refurbishment of the Kilbarry Marina 110 kV double circuit strengthens bulk supply to Cork city. These upgrades reflect a broader programme of reinforcement under EirGrid’s Transmission Development Plan, ensuring the physical network evolves alongside national targets [6].

Strategically, EirGrid and SONI have also published the All-Island Grid Forming Strategy [1], outlining how grid-forming inverter technologies will support system strength, inertia provision, and voltage stability as synchronous generation declines.

However, despite this progress, Ireland’s power system remains a physics-bound AC transmission network. Major demand centres remain concentrated in the east, while significant wind resources lie in the west and increasingly offshore. As new projects such as Tonn Nua (900 MW) and other south-coast developments advance toward planning and auction stages, the location and timing of generation injections become increasingly material to corridor loading and congestion patterns [3], [7].

As renewable penetration rises, operational constraints become more binding. The SNSP ceiling preserves frequency stability and system strength in a grid increasingly dominated by inverter-based generation [1]. Transmission corridors can experience congestion during high-wind periods [3], [7]. Interconnectors provide critical flexibility but are bounded by both physical capacity and market scheduling timeframes [4]. Together, these realities mean that additional offshore capacity must be integrated within an already tightly managed operational envelope.

The integration challenge is not capacity deployment, but constraint-aware coordination. The question that remains is how operational intelligence can evolve in parallel - how the system can anticipate constraint interactions, forecast stress conditions, and coordinate dispatch decisions before stability thresholds bind.

Digital Wind is introduced within this context as a grid-informed digital twin designed to enhance integration intelligence rather than replace infrastructure. Built upon real quarter-hourly system data, it first diagnoses how and why curtailment emerges under current operating conditions. It then stress-tests the network under incremental offshore expansion scenarios aligned with national targets, explicitly recognising transmission structure and grid constraints.

3. Policy & Regulatory Context

Ireland’s offshore wind expansion is embedded within a binding national and European decarbonisation framework. Under the Climate Action Plan and National Energy and Climate Plan (NECP), the State has committed to achieving 80% renewable electricity generation by 2030, supported by at least 9 GW of onshore wind and 5 GW of offshore wind capacity [1], [13]. These objectives align with the European Green Deal and EU-wide commitments to reduce greenhouse gas emissions by at least 51% by 2030 and achieve climate neutrality by mid-century [2]. These targets are legally anchored and carry financial and reputational implications if not delivered, reinforcing the urgency of effective renewable integration.

Current projections illustrate the delivery challenge. Under the “With Existing Measures” (WEM) scenario, Ireland is expected to achieve a renewable energy share of approximately 30.9%, significantly below the national target of 43%. Under the “With Additional Measures” (WAM) scenario - assuming full implementation of planned interventions—the projected share rises to 42.7%, narrowly below the target threshold [12] (Figure 1).

Parameter / Variable	Year	CAP24	WEM	WAM	Risk
RES-E (%)	2025	50%	46.4%	47.5%	46%
Onshore Wind Capacity (GW)	2025	6	5.6	5.8	5.2
Offshore Wind Capacity (GW)	2025	n/a	0.03	0.03	0.03
Solar PV Capacity (GW)	2025	5	2.2	2.2	2.2
RES-E (%)	2030	80%	68.9%	80.1%	54.7%
Onshore Wind Capacity (GW)	2030	9	6.8	7.2	6.2
Offshore Wind Capacity (GW)	2030	5	2.7	4.0	0.03
Solar PV Capacity (GW)	2030	8	5.7	6.5	5.1

Figure 1

Offshore wind development operates within a complex planning and regulatory environment. Projects must navigate maritime spatial planning requirements, environmental impact assessments, seabed leasing, fisheries consultations, and potential judicial review processes through An Coimisiún Pleanála and related authorities [14], [15], [16]. These procedural stages can significantly extend project timelines.

On the transmission side, EirGrid holds responsibility for maintaining system security, stability, and reliability across the all-island network. The system is undergoing structural reform to accommodate high levels of non-synchronous generation, with SNSP limits gradually increasing through system services innovation and operational reform [1]. Major infrastructure projects - including the Celtic Interconnector to France [10], the Greenlink interconnector to Wales [1], and the North–South Interconnector [17] - form part of a coordinated strategy to enhance security of supply, improve cross-border flexibility, and strengthen renewable absorption.

Policy also places increasing emphasis on demand-side flexibility and storage. The Climate Action Plan sets short-term flexibility targets of 15–20%, with longer-term goals reaching up to 50% [1]. Parallel reforms under the Future Arrangements for System Services (FASS) and related market redesign initiatives aim to incentivise fast-response technologies, grid-forming capabilities, and enhanced balancing services.

Digital Wind is positioned within this policy context as a complementary framework. It does not change regulatory processes or replace market mechanisms. Instead, it provides system-level insight into how offshore expansion interacts with transmission constraints.

4. Irish Power System Context & Fundamentals

Ireland operates one of the most advanced high-renewable electricity systems in the world. The all-island network, jointly operated by EirGrid and SONI, functions as a synchronised Alternating Current (AC) system serving demand that typically fluctuates between 2,500 MW and 7,000 MW [4]. As a relatively small and electrically isolated grid with limited synchronous coupling to the rest of the European continent, Ireland must manage stability margins with operational discipline.

4.1 AC System Fundamentals

The Irish grid operates through power flows governed by network physics, distributing across the system according to impedance relationships defined by Kirchhoff's laws [8].

Kirchhoff's Current Law states that the total current entering a node must equal the total current leaving it, while Kirchhoff's Voltage Law states that the sum of voltage differences around any closed loop must equal zero.

$$\sum I_{in} = \sum I_{out} \qquad \sum_{k=1}^m V_k = 0$$

- I_{in} : currents flowing into the node
- I_{out} : currents flowing out of the node
- V_k = the voltage drop (or rise) across the k^{th} element in a closed loop

Two forms of power are relevant in AC systems: active power (MW), which performs useful work, and reactive power (MVar), which supports voltage stability [8]. Renewable generation, particularly inverter-based wind power, contributes differently to system inertia and voltage control compared to conventional synchronous generators. As renewable penetration increases, maintaining frequency stability, voltage regulation, and system strength becomes more complex [4].

The prototype developed alongside this proposal does not explicitly model reactive power dynamics or voltage stability phenomena. Instead, it operates within the layer of active power balancing, focusing on real-power dispatch interactions that determine curtailment under high renewable penetration.

By forecasting offshore generation profiles and simulating their interaction with existing Irish grid constraints, *Digital Wind* estimates when and where operational limits would be approached. It then evaluates pre-emptive export and storage coordination strategies within defined interconnector capacities, ramp-rate limits, stability envelopes, and the pre-scheduled cross-border positions determined by day-ahead and intraday market commitments.

The digital twin therefore functions as a constraint-aware active power optimisation layer embedded within the physical boundaries of the AC transmission network underpinning the current Irish grid.

4.2 System Non-Synchronous Penetration (SNSP)

A defining operational parameter of the Irish grid is the System Non-Synchronous Penetration (SNSP) limit. SNSP represents the proportion of electricity supplied by non-synchronous sources - primarily wind, solar, and imported power delivered through HVDC interconnectors - that is relative to total system demand [4]. Since these sources are connected through power electronics rather than large synchronised rotating generators, they do not naturally damp sudden changes in system frequency. As SNSP increases, the grid becomes more sensitive to disturbances unless additional stability measures are actively maintained [4].

EirGrid has progressively raised the operational SNSP ceiling in recent years, reaching approximately 75% under secure conditions [4]. This ceiling defines a stability limit within which renewable integration must occur.

During periods of high wind and moderate demand, SNSP can approach this threshold. When the limit is reached, additional non-synchronous generation cannot be accommodated within secure operating margins, and wind output is reduced accordingly [5]. Offshore expansion therefore operates against a predefined stability ceiling that must be respected as the system currently exists.

In the *Digital Wind* framework, observed SNSP levels are treated as an explicit operational constraint within the predictive layer, enabling the model to anticipate when the stability ceiling is likely to bind and proactively coordinate export, storage dispatch, and other flexibility actions before curtailment occurs.

4.3 Transmission Infrastructure

The Irish transmission network operates across three principal voltage levels - 110 kV, 220 kV, and 400 kV. While 110 kV substations serve as regional transmission nodes, the 220 kV and 400 kV corridors form the backbone of primary power transfer across the island [3].

Demand is concentrated primarily around Dublin and the east coast, whereas a large share of wind generation is in the west and south-west [7]. This geographic imbalance introduces structural congestion risk, particularly during high-wind, low-demand conditions in export-constrained regions. The current *Digital Wind* prototype reflects this spatial structure by modelling offshore injections at high-voltage transmission nodes and evaluating export coordination within the limits of existing 220 kV and 400 kV backbone corridors on the east coast.

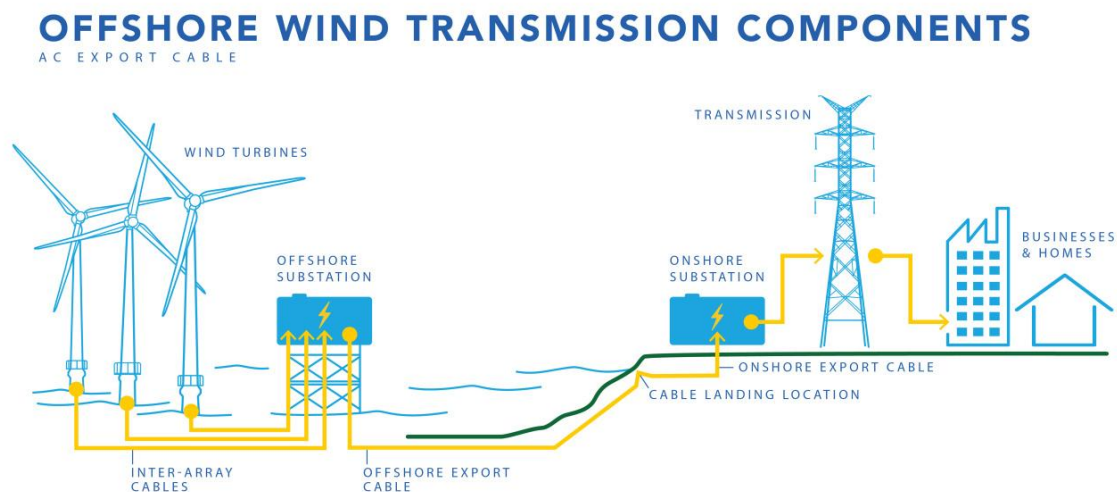


Figure 2

Figure 2 illustrates the simplified offshore transmission pathway and its integration into the high-voltage backbone corridors referenced above.

4.4 Interconnectors and Cross-Border Flows

Ireland operates High Voltage Direct Current (HVDC) interconnectors linking to Great Britain and, increasingly, continental Europe [3]. The Moyle interconnector (approximately 450 MW) connects Northern Ireland to Scotland, while the East–West Interconnector (approximately 529 MW) links Ireland to Wales. The Celtic Interconnector to France (700 MW) and the Greenlink interconnector to Wales further expand cross-border transfer capability [10], [11].

These HVDC interconnector's function both as security-of-supply assets and renewable balancing channels, enabling exports during wind surplus and imports during deficit conditions. However, cross-

border flows are not determined purely by real-time system need. They are shaped by day-ahead auctions and intraday market commitments, meaning that available transfer capacity at any given moment reflects both physical limits and pre-scheduled trading positions [9].

Digital Wind explicitly incorporates these interconnector realities within its predictive layer. The model treats interconnector capacity as both a physical constraint (maximum transfer limits and ramp-rate restrictions) and as a variable influenced by prior market decisions. By forecasting curtailment risk ahead of day-ahead and intraday gate closure, the system evaluates whether export positions can and should be adjusted within allowable limits before flows become fixed. Interconnectors are therefore not assumed to be freely dispatchable release valves; they are modelled as bounded flexibility assets whose effectiveness depends on anticipatory coordination rather than real-time reaction.

Being grounded in this operational reality, this proposal treats offshore expansion as a constrained optimisation challenge rather than a pure forecasting problem. The following stages quantify observed system behaviour and curtailment patterns before stress-testing future offshore injections and introducing a predictive decision-support framework.

5. Empirical Constraint Diagnosis and Curtailment Decomposition

5.1 Empirical System Baseline 2025 (Stage 1)

Stage 1 establishes an empirical baseline of Irish grid behaviour prior to introducing synthetic offshore injections. Rather than simulating an abstract network, *Digital Wind* is calibrated against observed 2025 operational data, ensuring that all subsequent stress testing occurs within a real and historic constraint context.

The analysis uses the official quarter-hourly operational dataset published by EirGrid and SONI, comprising 35,040 intervals across Ireland (IE), Northern Ireland (NI), and the All-Island (AI) system [18]. Core variables include:

- Generation and demand (IE, NI, AI)
- Wind and solar availability vs dispatched output
- SNSP (System Non-Synchronous Penetration)
- Interconnector flows (Moyle, EWIC)
- Oversupply indicators
- Inter-jurisdictional flows

5.1.1 System Scale and Renewable Contribution

Analysis proved the following results on renewable energy contributions across 2025:

- Total AI Demand (2025): 41,716 GWh
- Total Wind Generation: 13,624 GWh
- Total Solar Generation: 1,182 GWh
- Energy-weighted renewable share: 35.5%

Although instantaneous renewable penetration can be high, annual contribution remains materially below long-term targets. Curtailment and integration limits therefore directly influence effective renewable delivery.

5.1.2 Demand and Variability Structure

Demand peaks occur in the late afternoon (16:00–18:00), while wind output exhibits no strong diurnal structure (Figure 3). This temporal mismatch structurally increases curtailment exposure during nighttime low-demand periods and high-wind, moderate-load conditions.

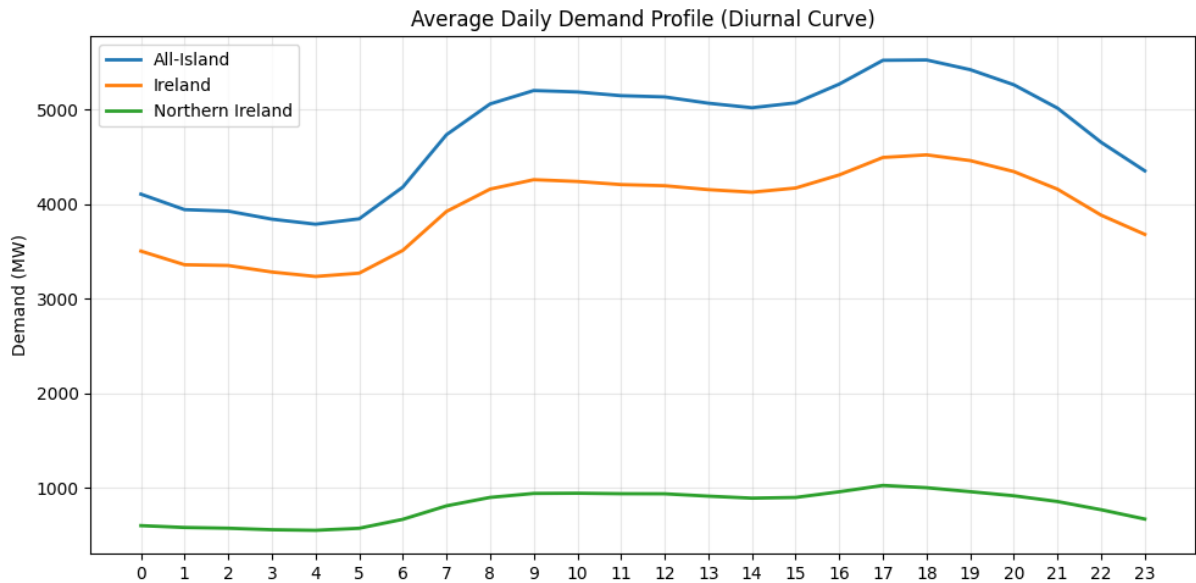


Figure 3

Solar energy generation remains comparatively small and does not materially drive system-wide curtailment events at present scale (Figure 4).

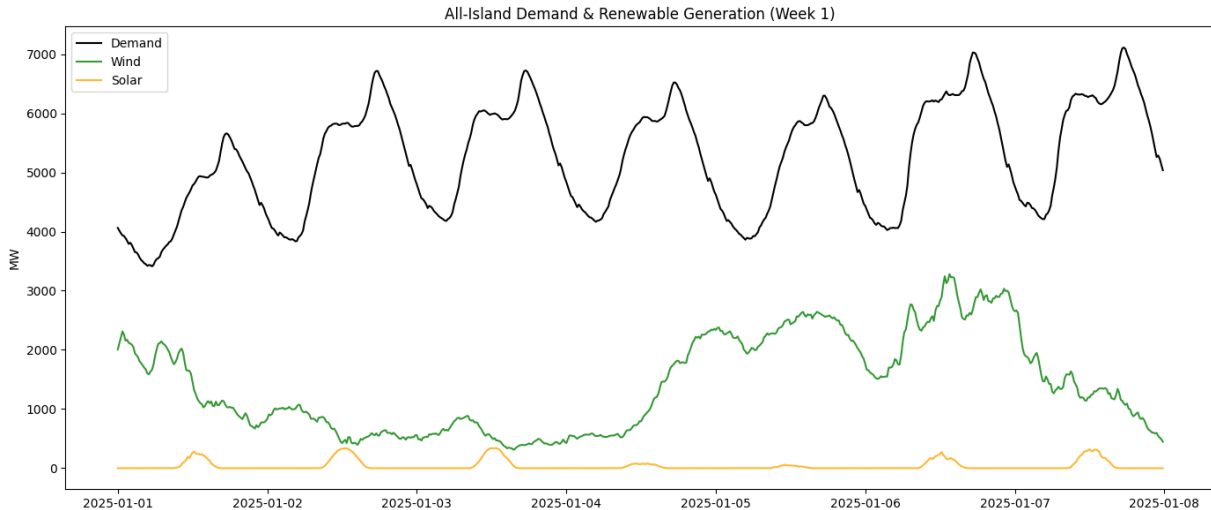


Figure 4

5.1.3 SNSP and Stability Envelope

The system was found to be frequently operating near its upper stability boundary ($\sim 75\%$), as shows in figures 5 and 6. When SNSP approaches this ceiling during high-wind events, wind dispatch must be reduced to maintain operational security. SNSP therefore functions as a binding operational constraint.

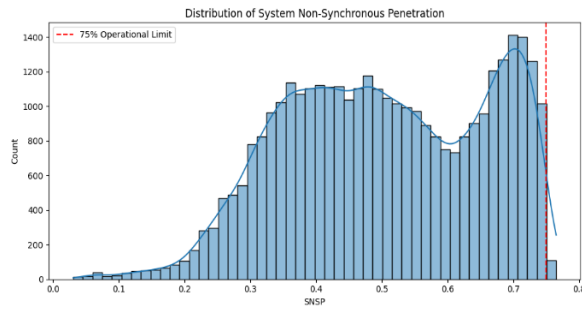


Figure 5

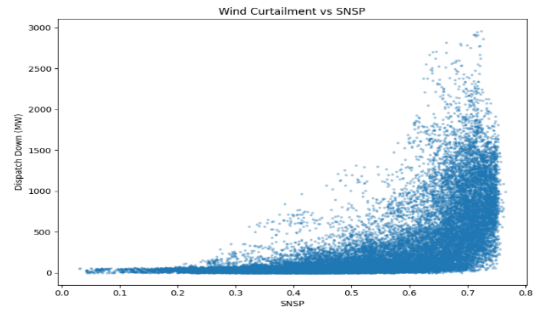


Figure 6

5.1.4 Curtailment Magnitude and System Envelope

Total wind dispatch-down (2025): 2,347.6 GWh, representing approximately 17% of available wind energy. A visual of total wind curtailed across the first week of 2025 is seen in Figure 7.

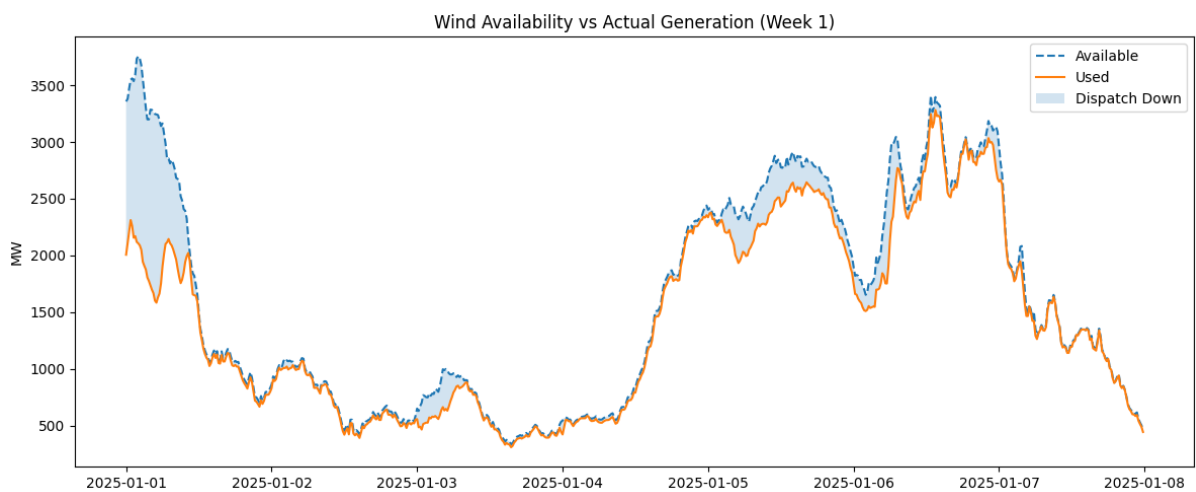


Figure 7

A representative stress interval occurred on 9 January 2025 at 17:30 UTC:

- Demand: 7,438 MW
- Wind: 141 MW
- Net demand: 7,297 MW

This low-wind, high-demand event illustrates that system stress occurs at both ends of the distribution: oversupply during high wind and adequacy risk during deficits.

Stage 1 therefore defines a dual-boundary operating envelope:

- Upper bound: SNSP and transmission congestion during high wind
- Lower bound: Net-demand stress during wind scarcity

Digital Wind is calibrated within this empirically observed envelope.

5.2 Curtailment Decomposition and Constraint Attribution 2025 (Stage 2)

While Stage 1 established when stress occurs, Stage 2 identifies the reason wind-generation is reduced. The official curtailment dataset (half-hourly aggregation) for the 2025-period [19] was utilized to decompose the dispatch-down (DD) regime into explicit causes:

$$DD = Constraints + Curtailments$$

- Constraints = Transmission constraints + TSO testing
- Curtailments = High-frequency/min-gen + RoCoF/Inertia + SNSP

Further analysis in stage 2 involved residual checks against the curtailment identities. Table 1 summarises the following:

	Check	Mean Residual (MWh)	Median Residual (MWh)	Max Abs Residual (MWh)	Non-zero %
0	DD - (CONSTRAINTS + CURTAILMENTS)	-0.000105	0.0	0.019	5.662100
1	CONSTRAINTS - (TRANS + TSO_TEST)	0.000000	0.0	0.000	0.000000
2	CURTAILMENTS - (HF + ROCOF + SNSP)	-0.000044	0.0	0.030	1.444064

Table 1

Residual means are effectively zero and maximum deviations are negligible (<0.03 MWh).

5.2.1 Annual Curtailment Structure

Aggregate wind reductions (MWh):

- Transmission constraints: 1,509.9
- High frequency / min generation: 664.1
- SNSP-specific reductions: 141.7
- RoCoF/Inertia: negligible

Transmission congestion is therefore the dominant structural driver of wind reduction in 2025. Curtailment is episodic rather than continuous: over half of intervals exhibit zero curtailment. In annual energy terms, transmission congestion dominates total curtailed volume. However, during high-SNSP intervals, stability-related reductions become proportionally more significant. Curtailment therefore exhibits regime-dependent causality.

5.2.2 Curtailment by SNSP Regime

Curtailment was decomposed by dominant cause across SNSP bands to examine how stability conditions influence constraint structure. Table 2 presents the percentage contribution of each reduction category within each SNSP regime.

DOMINANT_CAUSE	HF_MIN_GEN_MWH	SNSP_MWH	TRANS_CONSTR_MWH	TSO_TEST_MWH
SNSP_REGIME				
Low SNSP (<50%)	4.9	1.7	93.4	0.0
Medium SNSP (50–65%)	11.2	0.7	88.1	0.0
High SNSP (65–75%)	31.6	5.8	62.6	0.0
Very High SNSP (>75%)	29.2	16.9	53.8	0.0

Table 2

At low SNSP (<50%), curtailment is overwhelmingly transmission-driven (>90%). As SNSP increases, stability-related reductions (HF/Min Gen and SNSP-specific curtailment) become more material. However, even in high and very high SNSP regimes, transmission constraints remain significant, demonstrating that curtailment is multi-causal rather than solely driven by stability limits.

5.2.3 Curtailment and Interconnector Regimes

Figure 8 presents total curtailed wind energy by reduction category. Transmission constraints account for the largest share of annual curtailed energy, substantially exceeding stability-driven reductions (SNSP, high-frequency/minimum generation). This confirms that network congestion, rather than stability binding alone, is the dominant structural driver of wind curtailment in 2025.

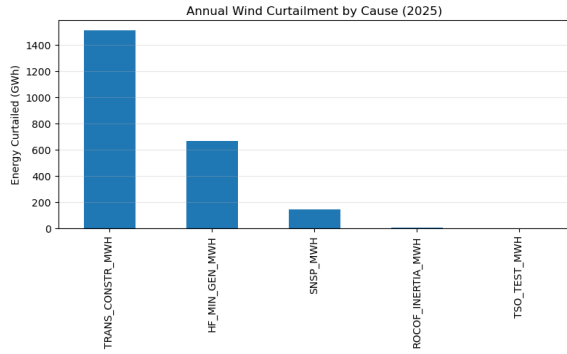


Figure 8

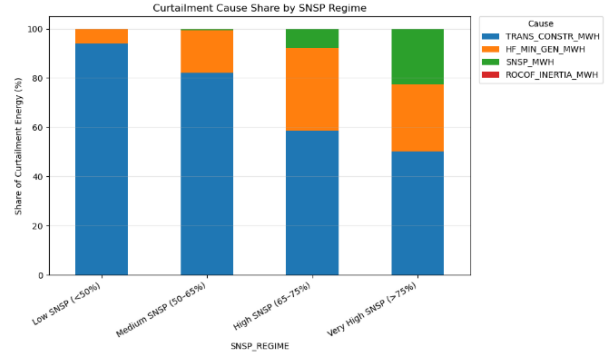


Figure 9

Figure 9 illustrates how the composition of curtailment evolves across SNSP bands. At low SNSP levels (<50%), transmission constraints dominate overwhelmingly. As SNSP increases toward the operational ceiling, stability-related reductions (high-frequency/minimum generation and SNSP-specific curtailment) become more material. Curtailment is therefore multi-causal and cannot be attributed solely to stability binding.

Collectively, these findings define the operational constraints of the 2025 Irish power system. Curtailment emerges from interacting transmission congestion, stability ceilings, and bounded interconnector flexibility rather than a single limiting factor. Offshore expansion must therefore be evaluated against this empirically observed system behaviour, not against simplified capacity assumptions. The *Digital Wind* framework builds directly on this quantified baseline, using these constraints to stress-test additional offshore injections and assess how predictive decision-making can improve renewable absorption within existing physical and market limits.

6. Offshore Wind Expansion Stress Testing (Stage 3)

6.1 Offshore Dataset and Profile Construction

Following the empirical diagnosis of the 2025 Irish system, Stage 3 introduces synthetic offshore wind injections to evaluate how additional capacity interacts with real-world transmission and stability constraints.

Offshore generation profiles were constructed using time-aligned historical wind generation and wind speed data from multiple operational European offshore wind farms [20]. The dataset comprises 350,640 high-resolution intervals and reflects real large-scale offshore performance characteristics.

Rather than injecting raw historical power directly into the Irish grid, offshore output was normalised using capacity factor (CF) [21], defined as:

$$CF_t = \frac{P_t}{P_{\text{rated}}}$$

- P_t = observed power output at time t ,
- P_{rated} = installed capacity of the source asset

Synthetic offshore injections were then constructed as:

$$P_t^{\text{offshore}} = CF_t \times C_{\text{scenario}}$$

- C_{scenario} = installed offshore capacity under each expansion scenario (0.5 GW, 1 GW, 5 GW).

This approach preserves realistic intermittency, ramp rates, and clustering of high-output events while allowing scalable deployment aligned with Ireland's offshore targets.

For each stress scenario, offshore output was generated as:

$$\text{Offshore MW} = \text{Capacity Factor} \times \text{Installed Capacity}$$

This ensures injections reflect real offshore performance rather than raw synthetic assumptions.

6.2 Spatial Transmission Injection Layer

Offshore generation was not uniformly added to every substation in the current Irish grid system (Figure 10). Instead, a spatial transmission layer was implemented to reflect realistic grid connection practice.

High-voltage substations (≥ 220 kV) along the east-coast area were identified and filtered to exclude non-landing infrastructure (highlighted in orange in Figure 11). The resulting offshore injection hubs were:

- Arklow 220 kV substation
- Carrickmines 220 kV substation
- Stepside 220 kV substation



Figure 10

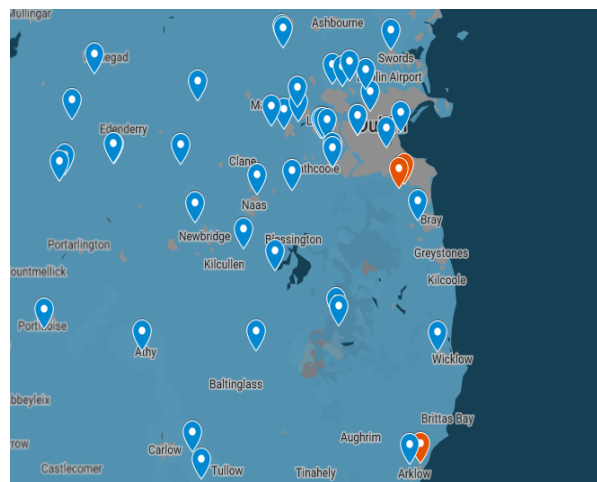


Figure 11

Each node was assigned a conservative thermal intake assumption of 500 MW, with a total structural intake capacity of 1.5 GW [22].

6.3 Offshore Capacity Scenarios

Three offshore expansion scenarios were evaluated:

- 0.5 GW - incremental deployment
- 1 GW - moderate expansion
- 5 GW - 2030 target-aligned offshore generation

At high penetration levels, transmission intake becomes the first binding constraint, independent of system-wide stability limits (Figure 12).

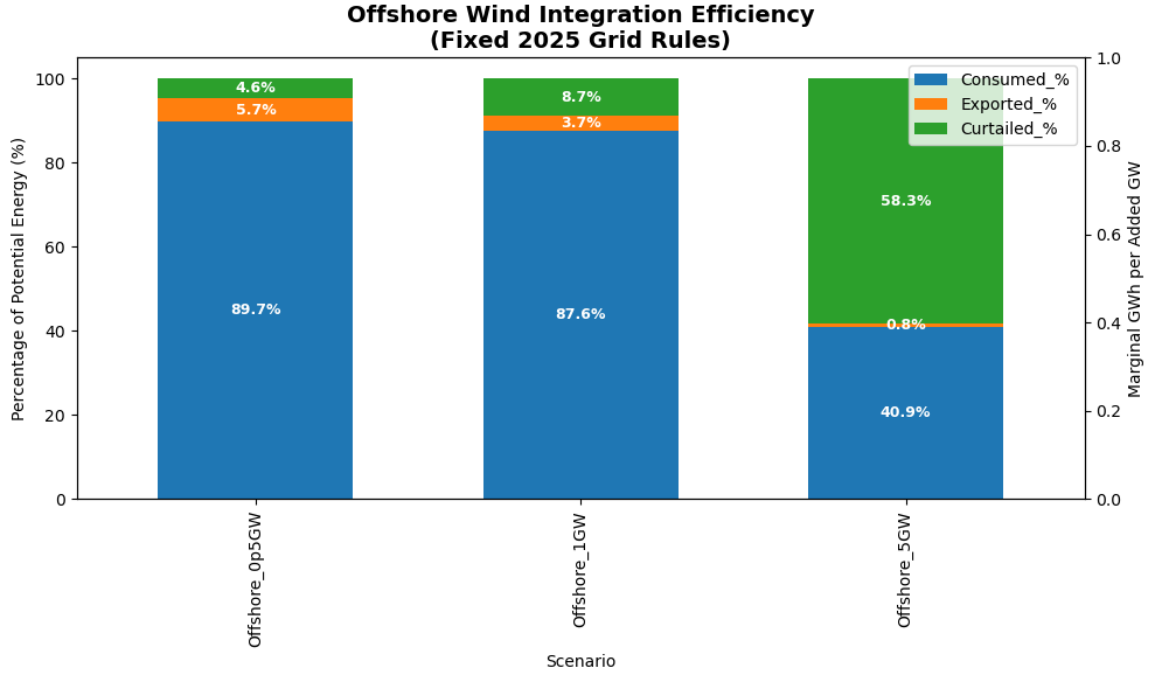


Figure 12

6.4 Constraint Re-Evaluation Framework

Following injection, constraints were re-evaluated sequentially:

1. **Structural Node Constraint**
Offshore output exceeding 1.5 GW intake was classified as local node congestion (Node Waste).
2. **Interconnector Export Headroom**
Accepted offshore output could be exported only if headroom existed in interconnector flow direction. Export capacity was bounded by observed cross-border flow conditions.
3. **SNSP Stability Constraint**
Total non-synchronous generation (wind + solar + offshore) was compared against the operational ceiling:

$$SNSP \leq 0.75 \times Demand$$

Excess non-synchronous generation beyond this threshold triggered system-level curtailment. Curtailment was applied first at the node level (thermal binding), then at the system level (SNSP binding), ensuring hierarchical realism within the grid system. Node-level intake binds prior to SNSP enforcement because it represents a local physical bottleneck independent of aggregate frequency stability. Only accepted offshore injection contributes to system-wide non-synchronous penetration.

6.5 Quantified Stress Results

Scenario	Consumed (GWh)	Consumed (%)	Exported (GWh)	Exported (%)	Curtailed (GWh)	Curtailed (%)	SNSP Binding Intervals	Primary Binding Constraint
Offshore_0p5GW	2,101	89.7%	134	5.7%	107	4.6%	813	Emerging SNSP exposure
Offshore_1GW	4,105	87.6%	172	3.7%	409	8.7%	1,284	Increasing SNSP binding
Offshore_5GW	9,583	40.9%	195	0.8%	13,651	58.3%	2,952	Structural intake + SNSP stability

Table 3

Table 3 summarises offshore energy routing outcomes under each expansion scenario within the 2025 system envelope. At 0.5 GW, offshore generation is largely absorbable within existing grid constraints, with minimal curtailment and limited SNSP binding exposure. At 1 GW, curtailment increases nonlinearly, reflecting growing interaction with the SNSP stability ceiling rather than structural intake limits. At 5 GW, more than half of offshore generation is curtailed under current grid conditions. Structural transmission intake limits bind first, and as accepted offshore generation increases system-wide non-synchronous share, SNSP stability constraints further restrict absorption. Together, these mechanisms define the integration ceiling of the present network configuration.

6.6 Structural Interpretation

The stress testing demonstrates three key dynamics:

- Curtailment scales nonlinearly with offshore capacity.
- Transmission intake limits bind before system-wide stability at high penetration.
- Interconnectors provide limited relief under large-scale expansion.
- SNSP binding intensifies as offshore share increases.
- Under a 5 GW expansion aligned with 2030 ambitions, existing infrastructure cannot absorb most of the additional offshore generation without coordinated intervention.
- A 58.3% curtailment rate under 5 GW indicates structural infeasibility within the current network configuration absent coordination or infrastructure reinforcement.

Stage 3 therefore quantifies the integration gap between 2030 offshore ambition and current operational capacity. This quantified stress exposure forms the input to Stage 4, where predictive coordination is introduced to mitigate curtailment within the existing constraint envelope.

7. Predictive Digital Twin - Decision-Support Layer (Stage 4)

The offshore wind injections designed in stage 3 quantified the integration gap between offshore expansion ambitions and the operational limits of the existing grid. The final component to this proposal is stage 4, which introduces a predictive coordination layer designed to operate strictly within that constrained environment. Rather than expanding infrastructure, this stage tests whether anticipatory decision-making alone can measurably reduce curtailment exposure.

The central construct is a predictive digital twin, implemented as a multi-model machine learning architecture embedded within a closed-loop dispatch simulation. The digital twin does not attempt to replicate every physical dynamic of the grid. Instead, it models the decision-relevant state variables and constraint signals required for anticipatory coordination across controllable flexibility channels, including interconnector export and import adjustment, conventional redispatch, and battery charging or discharging actions that absorb excess wind during forecast constraint periods and release it when system headroom returns.

7.1 Decision-State Formalisation

Let the system state at time t be represented by the vector:

$$\mathbf{x}_t \in \mathbb{R}^n$$

where $\mathbf{x}_t$ includes:

- Offshore generation
- System demand
- SNSP level
- Interconnector flows
- Curtailment levels
- Transmission congestion indicators
- Derived stress metrics

For each offshore scenario $s \in \{0.5, 1, 5\}$ GW, a scenario-specific state table is constructed. All features are chronologically aligned, and leakage-prone variables are excluded.

Leakage-prone variables include any features that directly or indirectly encode future information relative to the forecast origin time t , such as:

- Future target columns (e.g., variables containing “ $t + h$ ” shifts),
- Realised future curtailment values.
- Realised future interconnector flows.
- Derived indicators computed using future outcomes (saturation flags defined using full-sample thresholds).

Formally, the developed digital twin model is trained to learn:

$$\mathbf{x}_t \rightarrow y_{t+h},$$

- $\mathbf{x}_t$ = the system state vector at time t .
- y_{t+h} = the operational outcome at a future horizon h hours ahead.

This ensures that the feature space reflects only information available at the current decision time, preventing artificially inflated predictive performance, which fulfils the objective to construct a decision-ready state representation that preserves operational realism while preventing forward information contamination.

7.2 Engineered Constraint Indicators

To improve predictive sensitivity, raw state variables are transformed into higher-order constraint signals to better reflect binding dynamics. Feature engineering included dimensionless stress indicators that measure *proximity to binding limits* rather than absolute magnitude.

$$IC_Stress_t = \frac{|IC_Flow_t|}{IC_Capacity}$$

$$Congestion_t = \frac{Node_Waste_t}{Offshore_MW_t}$$

- IC_Stress_t = Interconnector stress ratio
- $Congestion_t$ = Congestion intensity

The interconnector stress ratio normalises cross-border flow relative to its physical limit. A value approaching 1 indicates imminent saturation, whereas values near 0 imply substantial headroom. This removes scale dependence and makes constraint pressure directly comparable across the three scenarios.

The congestion intensity ratio expresses the proportion of offshore generation that is effectively unusable due to network bottlenecks. Rather than learning from absolute curtailed megawatts - the models learn the *fractional constraint severity*.

These transformations convert physical quantities into regime indicators. Instead of inferring constraint dynamics indirectly from raw MW values, the learning algorithm receives explicit signals describing how tightly the system is operating against structural limits. As a result, the predictive layer becomes sensitive to stress states and transition boundaries.

7.3 Multi-Horizon Forecast Framework

The evaluated forecast horizons $h \in \{4, 24\}$ hours were chosen to explicitly represent intraday (4h) and day-ahead (24h) operational foresight, reflecting the structure of European electricity markets and cross-border scheduling mechanisms, where interconnector flows are partially pre-committed through day-ahead and intraday market decisions. Once nominated, these cross-border positions constrain the physical flexibility of the system.

- **24-hour forecasts** align with day-ahead market clearing. At this stage, cross-border capacity allocations and energy schedules are largely fixed. Poor anticipation of high-wind periods can result in insufficient export positioning, leading to structural curtailment.
- **4-hour forecasts** capture near-real-time balancing dynamics. At this stage, actions are primarily constrained by ramp limits and residual headroom rather than market nomination freedom.

As export flows cannot be freely adjusted at all timescales, inaccurate forward positioning can lock the system into suboptimal cross-border configurations. Under high offshore penetration, this may result in wind being curtailed despite available external demand, purely due to prior scheduling decisions.

Chronological splits in machine learning model training enforce:

$$Time_{train_data} < Time_{test_data}$$

referring strictly to temporal ordering. All observations used for training occur earlier in time than those used for evaluation.

In this prototype, a 70/30 split is applied for regression models and an 80/20 split for classification models, with the earlier 70–80% of the time series used for training and the most recent 20–30% reserved for out-of-sample testing.

This structure prevents look-ahead bias, ensures that performance metrics reflect genuinely unseen future periods, and mirrors real-world operation where forecasts must be generated using only historically available information within day-ahead and intraday market commitment frameworks.

7.4 Multi-model Machine Learning Architecture

The digital twin is structured as a closed-loop predictive layer comprising multiple interconnected sub-models. The output of each component informs the subsequent stage, creating a sequential decision architecture rather than a set of independent predictors.

7.4.1 Offshore Generation Forecast

A gradient boosting regression model [23] is trained to estimate future offshore wind generation at horizon $t + h$ using only the observable system state at time t :

$$\hat{G}_{t+h} = f_G(\mathbf{x}_t)$$

A chronological 70/30 split is applied, with the earlier 70% of observations used for training and the most recent 30% reserved for out-of-sample testing. The model is implemented as a Gradient Boosting Regressor, which builds an ensemble of shallow decision trees sequentially, each correcting residual errors from the previous stage.

The trained model (as well as every subsequent model) along with its preprocessing metadata (feature lists and encoders), is stored as a deployable bundle so that new real-time state vectors can be transformed and forecasted consistently during simulation.

7.4.2 Curtailment Event Probability

When expected wind output increases, the likelihood of constraint binding rises. Curtailment is therefore modelled as a probabilistic event driven by supply–constraint imbalance.

A logistic regression classifier [24] estimates the probability that curtailment will occur at the forecast horizon:

$$\hat{p}_{t+h} = P(C_{t+h} > 0 \mid \mathbf{x}_t)$$

- $\mathbf{x}_t$ = the observable system state at time t ,
- C_{t+h} = realised curtailment at horizon h .

The model uses current offshore generation levels, demand, interconnector flows, SNSP indicators, and congestion metrics to estimate whether the system will enter a binding regime. The output is a predicted probability rather than a binary decision, allowing the digital twin to quantify forward curtailment risk under anticipated wind and constraint conditions.

7.4.3 Curtailment Magnitude (Conditional)

While the previous model estimates whether curtailment will occur, it does not quantify how severe the event will be. Once the probability of entering a binding regime is established, the digital twin's next step is to estimate the expected scale of that constraint.

A regression model therefore estimates curtailment magnitude conditional on an event occurring:

$$\hat{M}_{t+h} = f_M(\mathbf{x}_t \mid C_{t+h} > 0)$$

- $\mathbf{x}_t$ = the system state at time t (the model being trained only on historical periods in which curtailment was strictly positive)

The curtailment magnitude model uses the same state variables - offshore generation, demand, SNSP level, interconnector flows, and congestion indicators - to estimate how strongly constraints will bind once its previous counterpart is triggered.

7.4.4 Expected Curtailment

The probability and magnitude models are combined to produce a single continuous exposure metric:

$$\hat{E}[C_{t+h}] = \hat{p}_{t+h} \times \hat{M}_{t+h}$$

- $\hat{p}_{t+h}$ = the predicted probability that curtailment occurs at horizon h ,
- $\hat{M}_{t+h}$ = the predicted magnitude of curtailment conditional on an event.

This formulation represents the expected value of future curtailment under the current system state. It transforms a binary event prediction into a continuous measure of anticipated wind loss. The probability model determines whether the system is likely to enter a binding constraint regime, while the magnitude model estimates the severity of that regime. Their product yields the expected curtailment exposure.

This output becomes the primary control signal in the digital twin. Rather than reacting to realised curtailment, the controller acts on forecast expected exposure, enabling anticipatory coordination within interconnector and operational limits.

7.4.5 Interconnector Flow & Saturation

To determine whether export support is physically feasible, additional models forecast future interconnector behaviour. A regression model estimates total interconnector flow at horizon h :

$$\hat{F}_{t+h} = f_F(\mathbf{x}_t)$$

These forecasts capture expected cross-border utilisation under current system conditions. An additional logistic regression model estimates the probability that interconnector flow will approach or exceed a predefined operational threshold:

$$P(|F_{t+h}| > F_{\text{threshold}})$$

- $F_{\text{threshold}}$ = a near-capacity stress level entity derived from historical training data.

These models are grounded in the operational reality of day-ahead and intraday electricity markets. Interconnector flows are not freely adjustable in real time; they are partially determined by prior market clearing outcomes, cross-border nominations, and scheduled exchange positions. As a result, available export headroom at time $t + h$ depends on commitments already made in earlier trading windows.

If projected flows are already near saturation due to prior market allocations, export-based mitigation is structurally constrained. If sufficient headroom is forecasted within the market commitment framework, anticipatory reallocation becomes operationally feasible.

7.5 Closed-Loop Digital Twin Simulation

The stacked predictive layer is embedded within a rolling, timestep-by-timestep simulation that emulates real operational decision-making. Rather than producing isolated forecasts, the digital twin continuously observes the system state, anticipates future constraint exposure, and applies corrective actions within physical and market limits.

At each timestep t , the simulator performs the following sequence:

1. **State Observation**

The current grid state $\mathbf{x}_t$ is observed. This includes offshore generation, demand, SNSP level, interconnector flows, congestion indicators, and other stress metrics.

2. **Forecast Exposure**

The stacked architecture computes expected curtailment at horizon h (if no intervention occurs):

$$\hat{E}[C_{t+h}] = \hat{p}_{t+h} \times \hat{M}_{t+h}$$

3. **Assess Feasibility Constraints**

Before any action is taken, the simulator evaluates operational headroom:

- SNSP stability margin
- Interconnector capacity limits
- Ramp-rate constraints on flow adjustment
- Forecast interconnector saturation risk

4. **Export Adjustment Decision**

If expected curtailment exceeds a predefined decision threshold τ , the controller attempts to increase interconnector export support. However, the adjustment is limited by:

- Maximum interconnector capacity
- Maximum allowable ramp per timestep
- Stability constraints

If forecast headroom is insufficient, no adjustment is made.

5. **Realised Mitigation Calculation**

The simulator compares adjusted export support against actual realised curtailment:

$$Wind_Saved_t = \eta \cdot \min(F_t^{support}, C_t^{actual})$$

- η = dispatch efficiency (how much wind energy would have been preserved under anticipatory coordination)

6. **State Update and Loop Continuation**

The system state is updated to $\mathbf{x}_{t+1}$, and the process repeats.

This structure constitutes a simplified Model Predictive Control (MPC) mechanism [25]. Unlike classical deterministic MPC, which optimises against explicit physical equations, this controller is driven by learned probabilistic forecasts of constraint exposure. The predictive models supply forward risk signals, while the control layer enforces hard operational limits.

Formally, the control objective can be expressed as:

Minimise:

$$J = \sum_t (C_t - Wind_Saved_t)$$

Subject to:

$$|F_t| \leq F_{max}$$

$$SNSP_t \leq 0.75 \times Demand_t$$

$$|\Delta F_t| \leq Ramp_limit$$

- C_t = realised curtailment,
- F_t = interconnector flow

The predictive layer supplies probabilistic estimates of constraint exposure, while the control layer enforces hard operational feasibility.

In reference to the full architectural diagram presented in Figure 13, the closed-loop control stage represents the final integration layer - connecting derived outputs, curtailment forecasts and interconnector feasibility predictions to physically constrained dispatch action, completing the end-to-end digital twin framework.

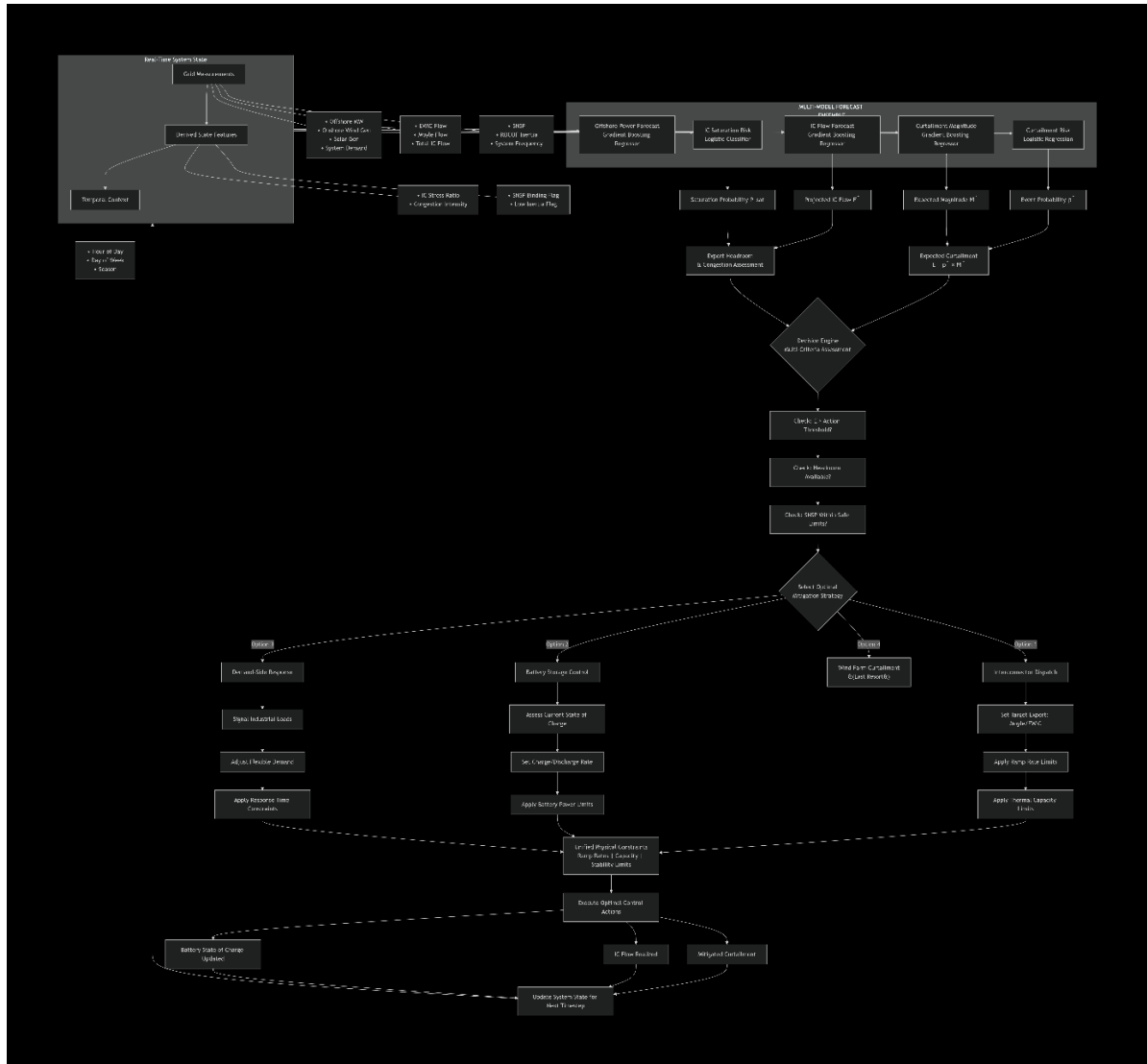


Figure 13

8. Results & Quantified Impact

The performance of each predictive component is summarised below. Results are reported across offshore penetration scenarios (0.5 GW, 1 GW, 5 GW) and forecast horizons.

8.1 Predictive Model Performance Summary

Model Component	Metric	Performance Range
Offshore Generation Forecast	RMSE (MW)	0.5 GW: 125–143 1 GW: 251–287 5 GW: 1,255–1,427
Curtailment Event Probability	ROC-AUC	0.55–0.70
Curtailment Magnitude (Conditional)	RMSE (MWh)	≈ 128 across all scenarios
Interconnector Saturation Risk	ROC-AUC (Short Horizon)	0.71–0.76

Table 4

The offshore forecast becomes progressively more difficult as penetration rises, consistent with increased volatility and ramp variability. Curtailment event prediction improves under higher offshore scenarios, reflecting clearer separation between non-binding and binding regimes. The magnitude model’s scale invariance suggests that once curtailment occurs, severity is dominated by structural constraints rather than installed capacity alone. Interconnector saturation forecasting provides the feasibility boundary required for anticipatory coordination (Table 4).

Importantly, predictive skill alone does not guarantee system benefit; its value emerges only when embedded within the closed-loop control framework. The observed mitigation arises not from perfect prediction accuracy, but from probabilistic anticipation within bounded flexibility. Even moderate predictive skill produces measurable benefit when applied ahead of binding constraint transitions.

8.2 Closed-Loop Mitigation Results

Scenario	Horizon	Wind Lost (MWh)	Wind Saved (MWh)	Net Mitigation (%)	Avg IC Support (MW)
0.5 GW	4 h	199,271	17,998	9.03	8.7
0.5 GW	24 h	198,760	15,367	7.73	8.6
1 GW	4 h	199,271	20,688	10.38	10.6
1 GW	24 h	198,760	19,327	9.72	10.6
5 GW	4 h	199,271	25,427	12.76	9.2
5 GW	24 h	198,760	21,847	10.99	11.1

Table 5

These results (Table 5) quantify the operational impact of embedding the predictive stack within a constraint-aware dispatch loop. Across scenarios, anticipatory export coordination reduces curtailment by approximately 7–13% without infrastructure expansion.

8.3 Structural Interpretation

Several structural patterns emerge:

1. **Mitigation scales with stress intensity.**
Higher offshore penetration yields larger absolute and relative mitigation. Under greater constraint pressure, anticipatory coordination has more leverage.
2. **Short-horizon control is more effective.**
Intraday (4h) forecasts produce stronger mitigation than day-ahead (24h), reflecting reduced forecast uncertainty and greater operational flexibility closer to real time.
3. **Efficiency over magnitude.**
Average interconnector support remains modest (~9–11 MW), indicating that mitigation arises from timing and positioning rather than large-scale export shifts.
4. **Constraint envelope remains binding.**
Even with predictive coordination, a significant portion of curtailment persists, confirming that structural infrastructure limits—not forecasting error alone—drive the integration gap.

9. Feasibility & Implementation Study

This study presents a proof-of-concept digital twin designed to operate within realistic transmission system operator (TSO) environments. The architecture prioritises deployability over algorithmic complexity. Models are computationally lightweight and suitable for real-time retraining on rolling operational datasets. 9.1 Computational Feasibility

The predictive layer is implemented using logistic regression classifiers and gradient boosting regressors. These algorithms are computationally efficient, well-understood, and suitable for structured tabular power system data.

Hyperparameter optimisation can improve marginal predictive performance; however, mitigation effectiveness is primarily determined by structural constraint modelling rather than algorithmic complexity. Parameters such as tree depth, learning rate, number of estimators, and regularisation strength may be tuned to enhance generalisation. This study prioritises deployability and operational realism over incremental gains in model accuracy.

9.2 Data Requirements

All datasets used in this study, along with preprocessing pipelines and model code, will be linked within the associated GitHub repository. This ensures reproducibility and transparency of both the modelling process and the closed-loop simulation.

9.3 Operational Integration Pathway

The digital twin can be integrated progressively rather than deployed as a wholesale system replacement.

9.3.1 Decision Support

The system operates as a forecasting dashboard, providing expected curtailment exposure, interconnector headroom projections, and stress regime indicators. This supports operator awareness without automated dispatch.

9.3.2 Market Positioning

Forecast outputs inform day-ahead and intraday interconnector nomination strategies. By anticipating high-wind periods and constraint exposure, cross-border positioning can be optimised before commitments are locked.

9.3.3 Closed-Loop Layer

The digital twin recommends ramp-constrained export adjustments within predefined operational boundaries. Actions remain subject to operator approval and institutional constraints.

Because the architecture overlays existing dispatch structures rather than replacing them, implementation risk is reduced. The predictive layer functions as a coordination enhancement, not a market re-clearing engine.

9.4 Scalability and Extension

The current implementation focuses on offshore wind and interconnector coordination at aggregated system level. However, the architecture is modular and extensible in several directions.

9.4.1 Algorithmic Enhancement

More advanced ensemble methods, calibrated probabilistic models, or hybrid physics-informed learning approaches could replace or augment the current regressors. Hyperparameter optimisation frameworks can further refine performance.

9.4.2 Turbine-Level Offshore Modelling

Rather than modelling offshore generation as an aggregate quantity, the framework could be extended to turbine-by-turbine or farm-level forecasting. This would enable:

- Wake interaction modelling

- Local congestion identification
- Region-specific curtailment risk assessment
- Floating wind variability integration

9.4.3 Battery and Storage Coordination

Storage systems can be introduced as additional control variables subject to state-of-charge and ramp constraints, expanding the mitigation envelope beyond export channels.

9.4.4 Reactive Power and Voltage Constraints

Incorporating reactive power flow and voltage headroom indicators would move the model closer to AC-constrained operation and enable local stability-aware coordination.

9.4.5 Solar Integration

Although this study focuses on offshore wind, the Digital Wind framework can be extended to accommodate growing solar penetration. Extending the current framework to include solar would enable coordinated management of various non-synchronous energy resources particularly in periods or regions where wind availability is low and solar generation provides a more stable supply profile.

Recent Irish events shows that pairing wind and solar reduces exposure to multi-day renewable energy source (RES) droughts. Moving from a wind-dominant electricity-generation toward a balanced wind-solar portfolio approximately halves the number of drought events and significantly reduces the total annual hours spent in low-output regimes [26].

However, diversification alters rather than eliminates system stress. Solar PV materially reduces summer drought exposure and smooths seasonal variability, but winter remains the critical adequacy period, when low renewable output can occur at the same time as extreme electricity demand.

Combining these findings within future development of *Digital Wind* would allow duration-based and seasonal stress modelling. Rather than forecasting aggregate renewable output alone, the models would quantify the probability, expect duration, and return period of low-output events under different wind-solar capacity configurations.

9.4.6 Geographic Expansion

The scenario framework can incorporate additional offshore development zones, including southern and Cork-based floating wind farm projects, enabling regionalised stress modelling [27].

This work demonstrates that predictive coordination can reduce curtailment within the existing Irish grid infrastructure. It does not attempt to simulate full AC network dynamics or redesign market clearing processes but represents a foundational step toward a broader grid-level artificial intelligence coordination platform capable of evolving alongside 2030–2050 decarbonisation ambitions.

10. Strategic Implementation Roadmap & Future Development

The roadmap can be viewed across three horizons:

- **Short-term (2030).**
Focus on export optimisation, strengthened intraday coordination, and measurable curtailment reduction within existing infrastructure. Predictive positioning improves utilisation of interconnector capacity before binding occurs.
- **Medium-term (2040).**
Integrate battery storage, reactive power modelling, and multi-node optimisation to address local voltage and congestion constraints. Expand regional modelling to accommodate additional offshore zones.

- **Long-term (2050+).**

Evolve toward a fully integrated digital grid platform managing wind, solar, storage, hydrogen production, and cross-border flows through coordinated artificial intelligence systems.

Predictive coordination complements infrastructure expansion by improving utilisation efficiency within existing constraints.

11. Conclusion

This study introduced a predictive digital twin for offshore wind integration, embedding multi-model machine learning models within a closed-loop, constraint-aware balancing simulator. The objective was to assess whether anticipatory coordination alone can reduce structural curtailment under increasing offshore penetration.

Results show that forecast-driven intervention strategies reduce curtailment by approximately 7–13% within the existing infrastructure. Short-horizon forecasts provide stronger leverage, and mitigation increases with penetration, indicating greater value under higher system stress.

The framework serves as a practical proof-of-concept. It uses lightweight, interpretable models, operates on existing operational data, and respects hard physical and institutional constraints. Curtailment reduction is achieved through predictive positioning rather than infrastructure expansion.

Digital Wind represents a constraint-aware probabilistic coordination layer embedded within an AC transmission system. A future platform would extend this architecture across wind, solar, storage, hydro and reactive power coordination. Anticipatory control does not eliminate the integration gap, but it measurably compresses it, strengthening Ireland’s ability to meet its 2030 renewable electricity targets while laying a scalable foundation for longer-term 2040–2050 decarbonisation objectives. By improving the efficiency of existing infrastructure and reducing structural curtailment, the framework contributes to enhanced energy security, system resilience, and more effective utilisation of national renewable resources.

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13. Appendices

13.1 Appendix A - AI Use Declaration

AI tools were used in a limited and assistive capacity during the preparation of this report. The following tools were used:

- ChatGPT (OpenAI)
- Gemini (Google)
- Claude (Anthropic)

These tools were used to support:

- Structural development and refinement of written sections
- Grammar correction and clarity improvements
- Formatting of citations and reference lists
- High-level research support (identifying relevant literature and sources)
- Debugging assistance and code clarification across four Python notebooks

All core intellectual contributions - including problem formulation, system modelling, feature engineering, dataset construction, machine learning implementation, simulation architecture, mathematical formulation, evaluation, and interpretation of results - were conducted independently by the author.

13.2 Appendix B - Team Contribution Statement

This project was completed independently. No team contributions apply.

13.3 Appendix C - Reproducibility Statement

Repository URL: <https://github.com/davidfit21/Digital-Wind-2026>

Commit hash used to generate reported results: 523a3f5b11207807e19481f06cf6887cd38c01f0

Branch used: main

Date of final experiment execution: 21 February 2026

Computational environment:

- Operating System: Windows 11
- Python version: 3.10.x

Data availability:

All datasets and preprocessing pipelines required to reproduce the reported results are either included directly within the repository (where file size permits) or linked to their official source locations in the repository README where dataset size exceeds GitHub limits.